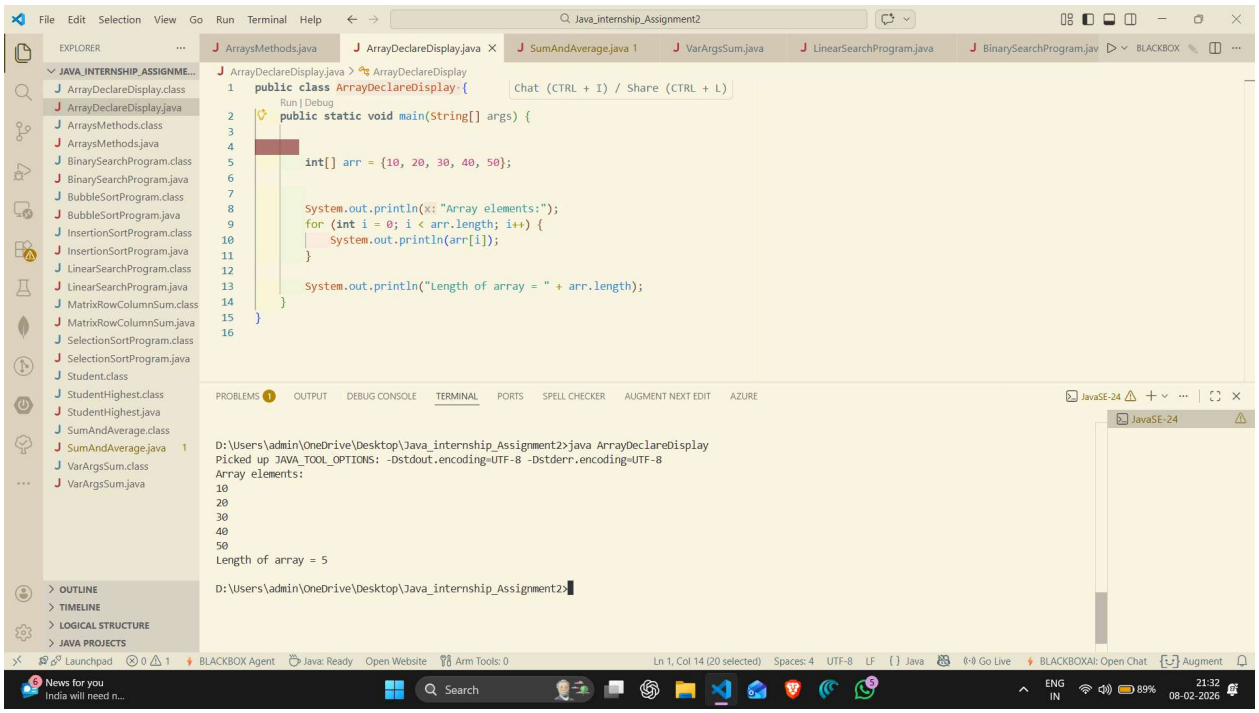
HARSHITH V-1MJ22CS073

MINDSET Java Programming Internship: Assignment-2(FEB-5 TO FEB-7,2026)

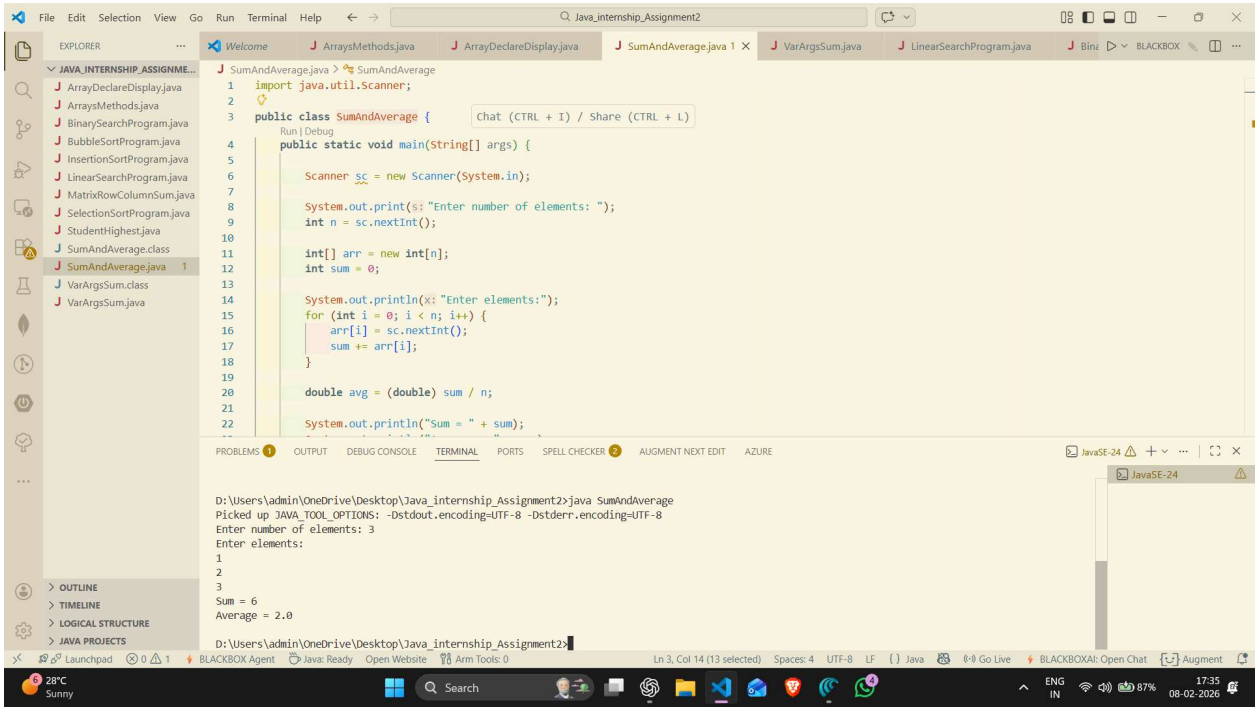
1. Array Declaration and Display

Concept: Demonstrates how to initialize a one-dimensional array and use a for loop to traverse and print its elements along with its length.



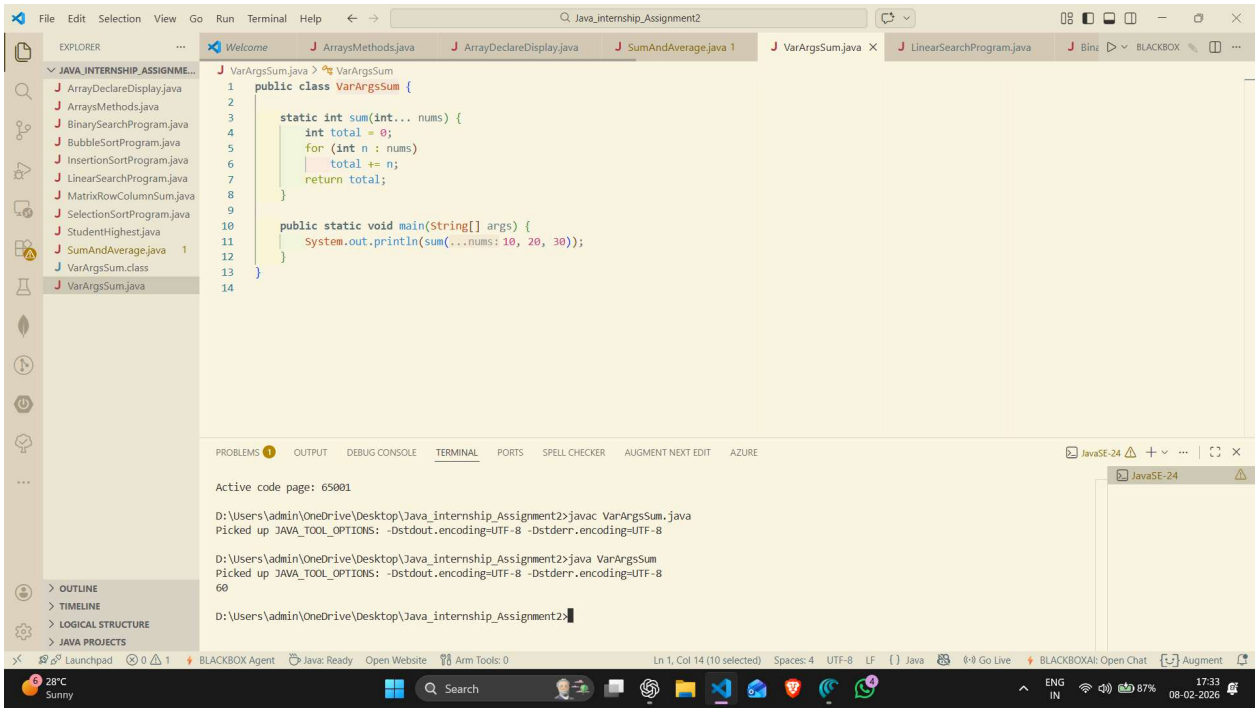
2. Sum and Average of Array Elements

Concept: Shows how to take user input for array size and elements using the Scanner class, then calculate the total sum and the average.



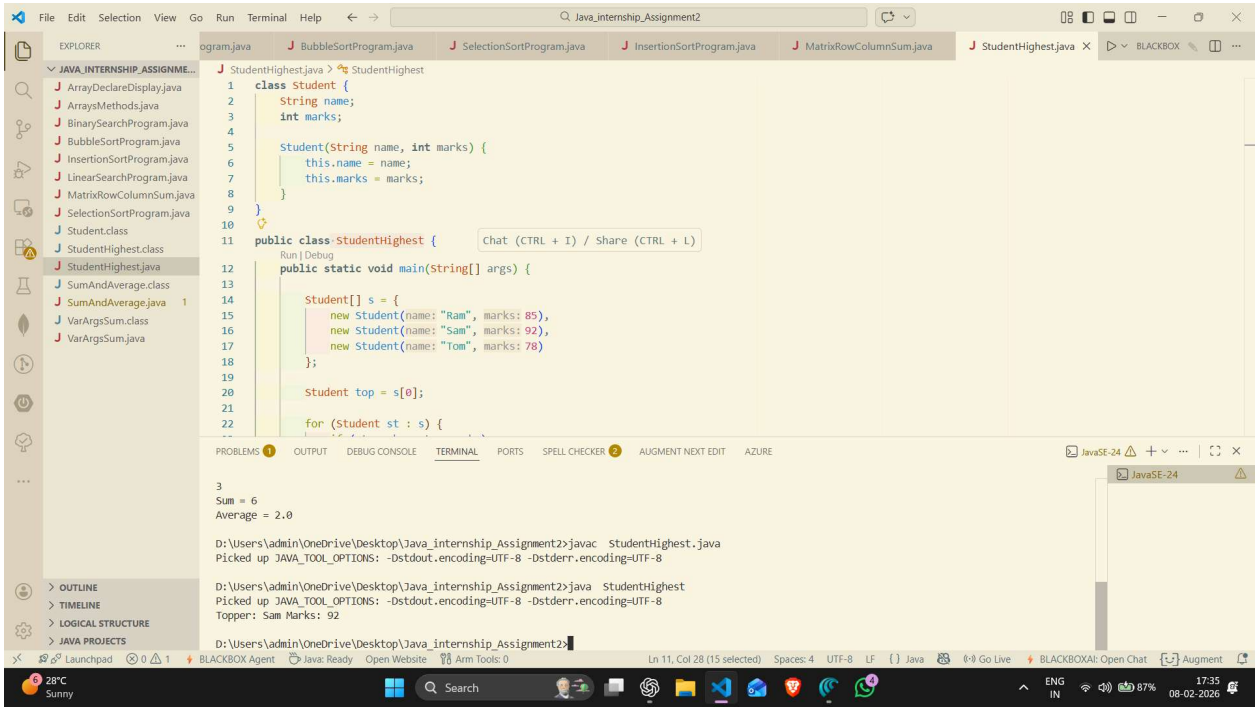
3. Variable Arguments (Varargs) Sum

Concept: Illustrates the use of the ... syntax (varargs) to create a method that can accept a variable number of integer arguments.



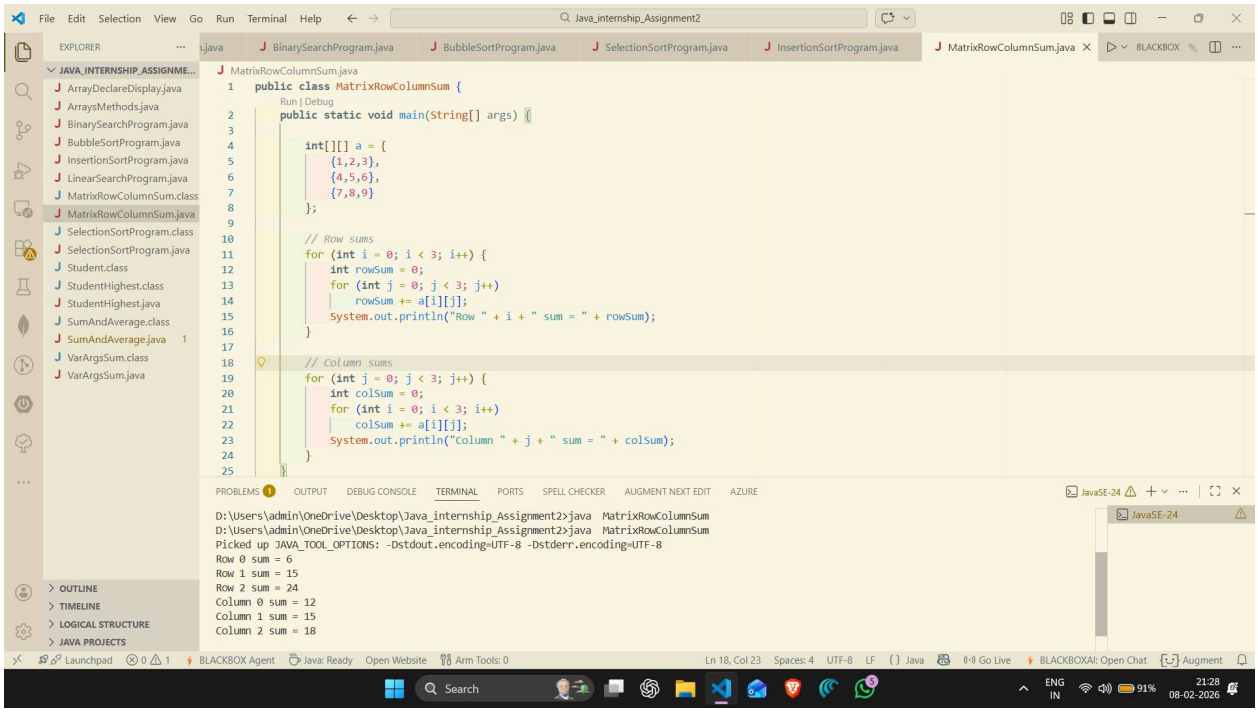
4. Student Array (Object Array)

Concept: Demonstrates how to create an array of objects and iterate through it to find the object with the highest value in a specific field.



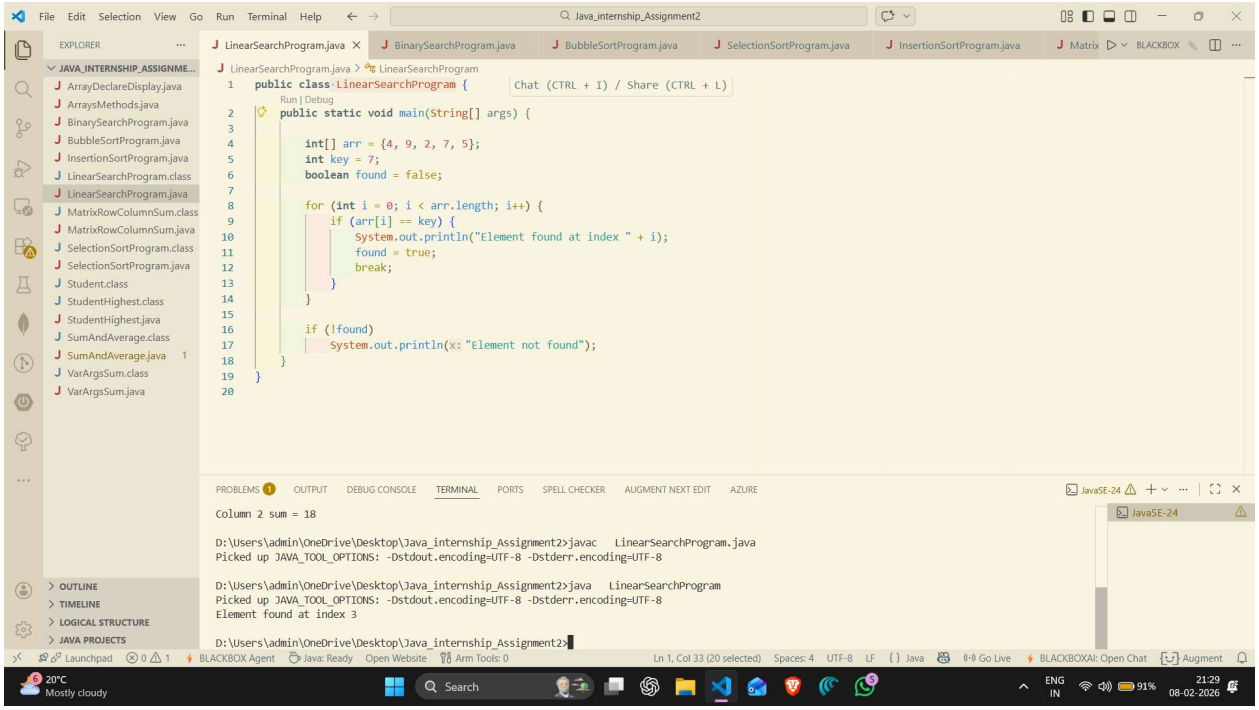
5. Matrix Row and Column Sum (2D Array)

Concept: Shows how to handle two-dimensional arrays by calculating the sum of each row and each column independently.



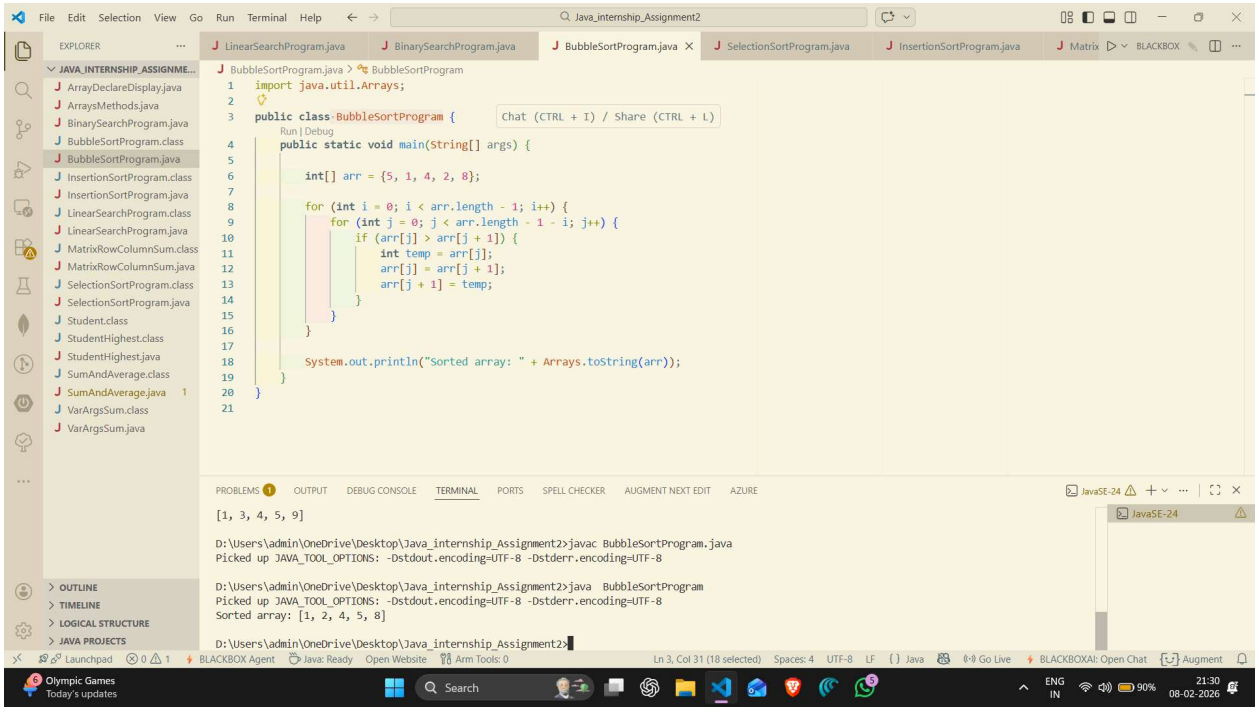
6. Linear Search Program

Concept: A basic search algorithm that checks every element in the array sequentially until the target key is found.



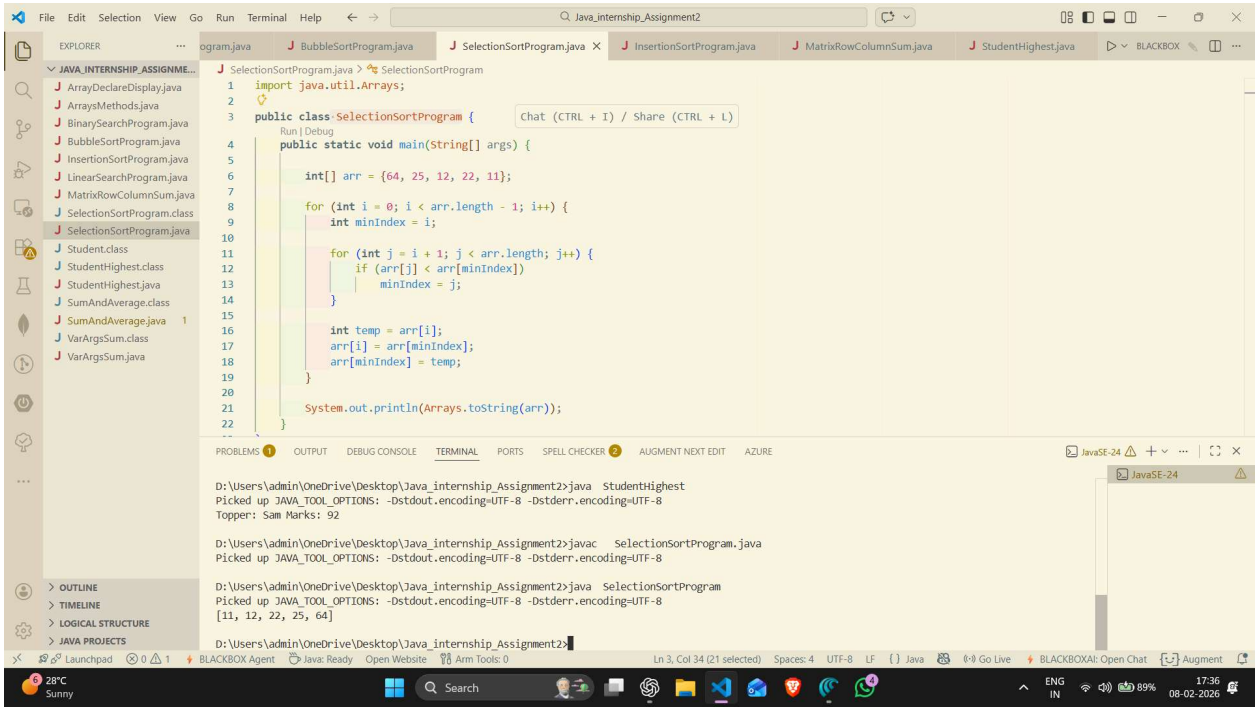
7. Bubble Sort Program

Concept: A sorting algorithm that repeatedly steps through the list, compares adjacent elements, and swaps them if they are in the wrong order.



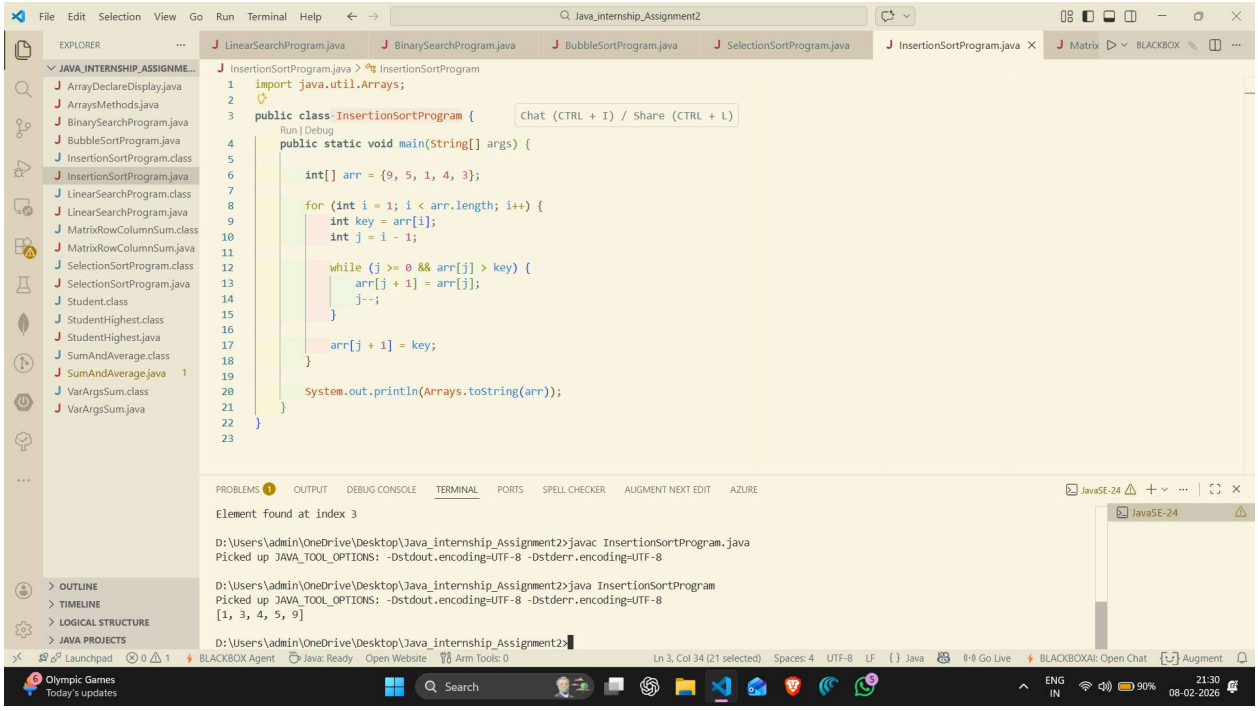
8. Selection Sort Program

Concept: An algorithm that divides the input list into a sorted and an unsorted sublist, repeatedly picking the smallest element from the unsorted sublist and moving it to the sorted sublist.



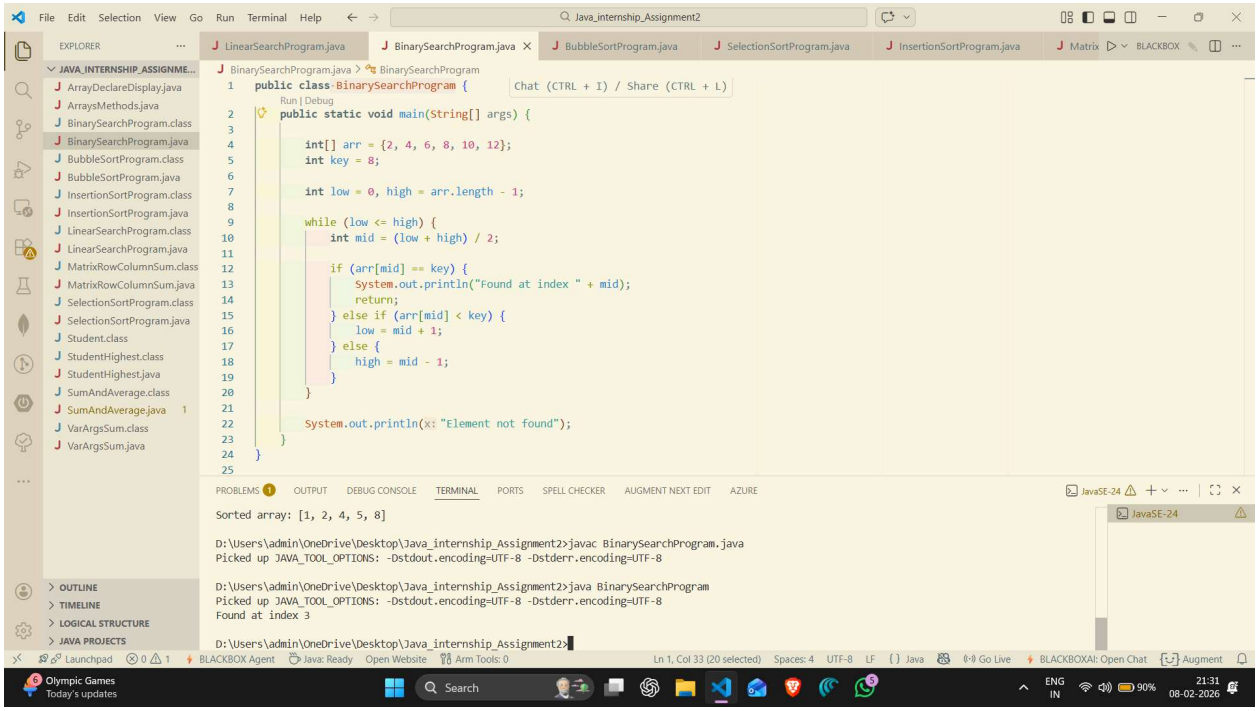
9. Insertion Sort Program

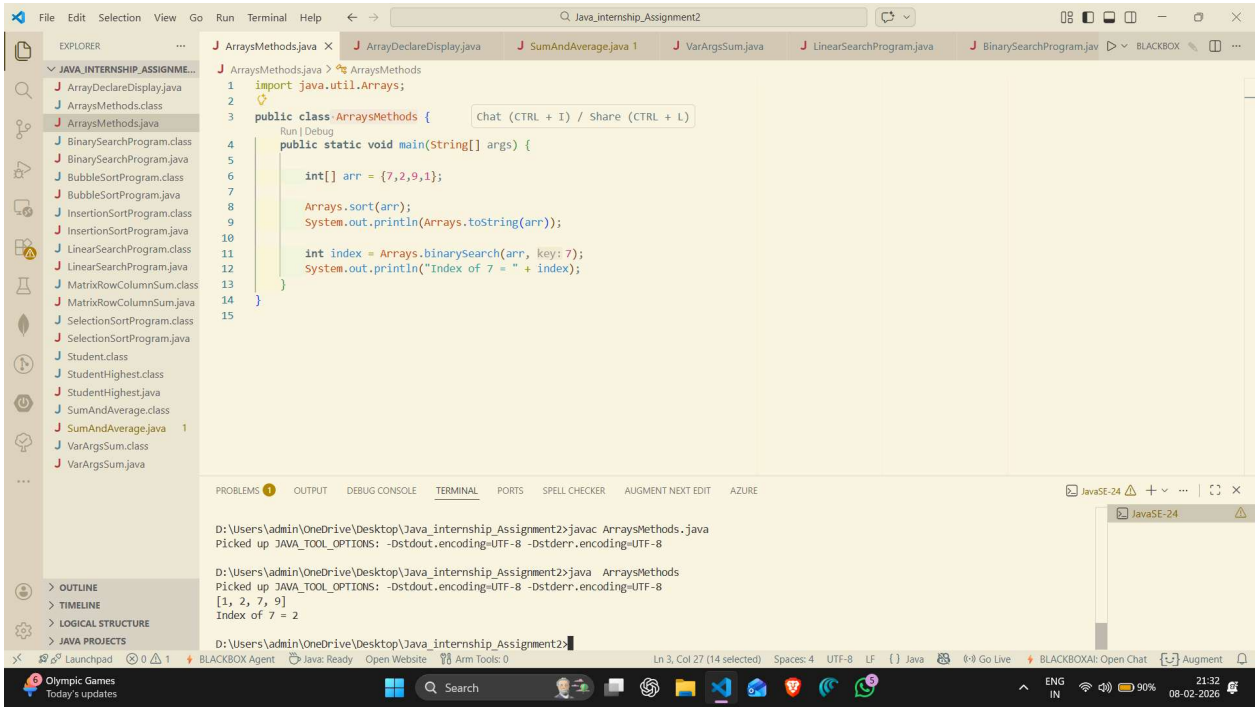
Concept: A sorting algorithm that builds the final sorted array one item at a time by consuming one input element each repetition and growing a sorted output list.



10. Binary Search Program

Concept: A highly efficient search algorithm that works on **sorted arrays**. It repeatedly divides the search interval in half by comparing the key to the middle element.



11. Built-in Array Methods

Concept: Demonstrates the use of Java's utility class `java.util.Arrays` to perform common tasks like sorting and searching without writing manual logic.